

Homework 3: Concise Answers

Formulae, tables, and diagrams

Games1 (p. 1)

(A) $x = 1, \dots, 9$: left to right; $y = 0, \dots, 9$: bottom to top. Red moves toward increasing y .

Piece	Position	Legal destination(s) / directional count	Count
Left rook	(1, 0)	(1, 1), (1, 2), (1, 3)	3
Right rook	(9, 0)	(9, 1), (9, 2)	2
Left horse	(2, 0)	(1, 2), (3, 2)	2
Right horse	(8, 0)	(7, 2), (9, 2)	2
Left elephant	(3, 0)	(1, 2), (5, 2)	2
Right elephant	(7, 0)	(5, 2), (9, 2)	2
Left advisor	(4, 0)	(5, 1)	1
Right advisor	(6, 0)	(5, 1)	1
General	(5, 0)	(5, 1)	1
Left cannon	(4, 2)	Left 3, right 3, down 1, up 6	13
Right cannon	(8, 2)	Left 3, right 1, down 1, up 4 noncaptures + 1 capture	10
Soldier 1	(1, 4)	(1, 5)	1
Soldier 2	(3, 3)	(3, 4)	1
Soldier 3	(5, 3)	(5, 4)	1
Soldier 4	(7, 3)	(7, 4)	1
Soldier 5	(9, 3)	(9, 4)	1

Right cannon capture: $(8, 2) \rightarrow (8, 9)$, with the single screen at $(8, 7)$.

$$b = 3 + 2 + 2 + 2 + 2 + 2 + 1 + 1 + 1 + 13 + 10 + 5 = \boxed{44}.$$

(B) One-ply leaf evaluations:

$$\boxed{44 \text{ heuristic evaluations}}.$$

(C) Three-ply leaf evaluations (constant b):

$$\boxed{b^3 = 44^3 = 85,184 \text{ heuristic evaluations}}.$$

If every non-root node is evaluated: $44 + 44^2 + 44^3 = 87,164$.

Game2 (p. 2)

Cell numbering:

1	2	3
4	5	6
7	8	9

X first; alternating moves. Two legal continuations:

First game. The supplied prefix is $X2, O5, X3, O1$. Add

$X9, O6, X4, O7, X8$.

O	X	X
	O	

 $\rightarrow$

O	X	X
	O	
		X

 $\rightarrow$

O	X	X
	O	O
		X

O	X	X
X	O	O
		X

 $\rightarrow$

O	X	X
X	O	O
O		X

 $\rightarrow$

O	X	X
X	O	O
O	X	X

Draw (no winning line).

Second game. The supplied prefix is $X2, O3$. Add

$X5, O8, X1, O9, X6, O4, X7$.

	X	O

 $\rightarrow$

	X	O
	X	

 $\rightarrow$

	X	O
	X	
	O	

 $\rightarrow$

X	X	O
	X	
	O	

X	X	O
	X	
	O	O

 $\rightarrow$

X	X	O
	X	X
	O	O

 $\rightarrow$

X	X	O
O	X	X
	O	O

 $\rightarrow$

X	X	O
O	X	X
X	O	O

Draw (no winning line).

Game3 (p. 3)

$U = +1$ (X win), -1 (O win), 0 (draw).

$$V(s) = \begin{cases} U(s), & s \text{ terminal,} \\ \max_{s' \in \text{children}(s)} V(s'), & X \text{ to move,} \\ \min_{s' \in \text{children}(s)} V(s'), & O \text{ to move.} \end{cases}$$

Terminal values:

$$\begin{array}{|c|c|c|} \hline X & O & X \\ \hline & O & X \\ \hline O & & X \\ \hline \end{array} +1 \qquad \begin{array}{|c|c|c|} \hline X & & X \\ \hline O & O & X \\ \hline O & & X \\ \hline \end{array} +1.$$

Upper fork:

$$\begin{array}{|c|c|c|} \hline X & X & O \\ \hline O & O & X \\ \hline X & O & X \\ \hline \end{array} 0 \qquad \begin{array}{|c|c|c|} \hline X & O & O \\ \hline O & O & X \\ \hline X & X & X \\ \hline \end{array} +1.$$

Upper fork: $\min(0, 1) = 0$; parent $\min(1, 0) = 0$. Single-successor chains retain the value.

Shared fork:

$$\begin{array}{|c|c|c|} \hline O & X & X \\ \hline X & X & O \\ \hline O & O & X \\ \hline \end{array} 0 \qquad \begin{array}{|c|c|c|} \hline O & X & X \\ \hline X & X & O \\ \hline O & X & O \\ \hline \end{array} +1.$$

Shared fork: $\min(0, 1) = 0$.

First X : corner. Pictured O replies:

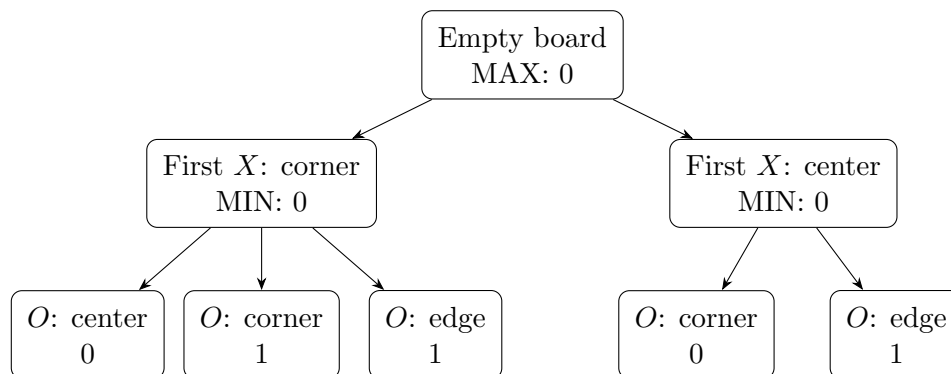
O in center : 0, O in top-right corner : 1, O on top edge : 1.

Dashed draw alternative at an X node: $\max(1, 0) = 1$.

Opening backup:

$$V(\text{corner first}) = \min(0, 1, 1) = 0, \quad V(\text{center first}) = \min(0, 1) = 0, \quad V(\text{root}) = \max(0, 0) = \boxed{0}.$$

Opening tree (single-successor chains compressed):



Choose center; corner ties at value 0.

Values apply to the supplied partial graph. In the full game, perfect play draws from every opening.

Game4 (p. 4)

Initial board:

		O	
		X	

$h(s) = \# \text{open } X \text{ lines} - \# \text{open } O \text{ lines}$. MIN over O replies:

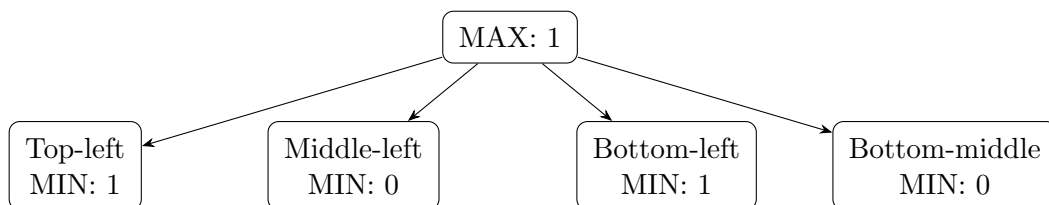
X 's candidate move	Scores after O 's responses	MIN value
Top-left corner	2, 1, 3, 1, 2, 1	1
Middle-left edge	1, 0, 2, 0, 1, 1	0
Bottom-left corner	2, 2, 3, 1, 2, 2	1
Bottom-middle edge	1, 1, 0 (symmetry representatives)	0

Root backup:

$$\max(1, 0, 1, 0) = \boxed{1}.$$

Best: all four corners (by symmetry).

Choose bottom-left as a tie-break: same worst-case value 1, more favorable other reply scores.



Chosen move:

		O	
		X	
X			

Game5 (pp. 5–6)

X first; root depth 0; stop immediately after a win or a full board. Leaves count move histories (no board or symmetry merging).

Item	Reason	Result
1	Third X mark: move 5; full board: move 9	Leaf depths 5, 6, 7, 8, 9
2	Depth 5: X has 3 marks, O has 2	$U_5 = +1$
3	Move 6 is O 's; earlier X wins already stopped	$U_6 = -1$
4	Move 9 is X 's: win or full-board draw	$U_9 \in \{0, +1\}$

5. Number of depth-5 leaves. 8 winning lines; $3!$ orders for X ; $\binom{6}{2}2!$ placements/orders for O .

$$N_5 = 8 \times 3! \times \binom{6}{2} \times 2! = 8 \times 6 \times 15 \times 2 = \boxed{1,440}.$$

6. Number of depth-6 leaves (optional). Count O wins, then exclude earlier X wins:

$$8 \times 3! \times \binom{6}{3} \times 3! = 5,760.$$

Disjoint ordered winning-line pairs: $3 \cdot 2 + 3 \cdot 2 = 12$. Invalid orders per pair: $3!3! = 36$.

$$N_6 = 5,760 - 12 \times 36 = \boxed{5,328}.$$

7. Depth-9 leaves (optional). $L_d(s, t)$ counts terminal continuations in exactly d further moves:

$$L_0(s, t) = \begin{cases} 1, & s \text{ is terminal,} \\ 0, & \text{otherwise,} \end{cases}$$

and for $d > 0$,

$$L_d(s, t) = \begin{cases} 0, & s \text{ is terminal,} \\ \sum_{a \in \text{legal}(s, t)} L_{d-1}(\text{Result}(s, a), \bar{t}), & \text{otherwise.} \end{cases}$$

Enumeration of the unmerged legal tree:

Depth	X wins	O wins	Draws	Terminal histories
5	1,440	0	0	1,440
6	0	5,328	0	5,328
7	47,952	0	0	47,952
8	0	72,576	0	72,576
9	81,792	0	46,080	127,872

Thus

$$\boxed{N_9 = 81,792 + 46,080 = 127,872}.$$

Check: 16 full draw boards; every mark order is legal:

$$16 \times 5! \times 4! = 46,080.$$

Total terminal histories:

$$1,440 + 5,328 + 47,952 + 72,576 + 127,872 = 255,168.$$

9! is invalid: it includes histories continuing after an earlier win.